INDIAN INSTITUTE OF TECHNOLOGY ROPAR

Department of Mechanical Engineering

Increasing the Life of Reusable Rocket Engines: Mitigating the Doghouse Effect at the Regeneratively Cooled Nozzle Throat using Pre-Domed cooling Channel Geometries

Presented by

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MTech Manufacturing Engineering

Project Guide

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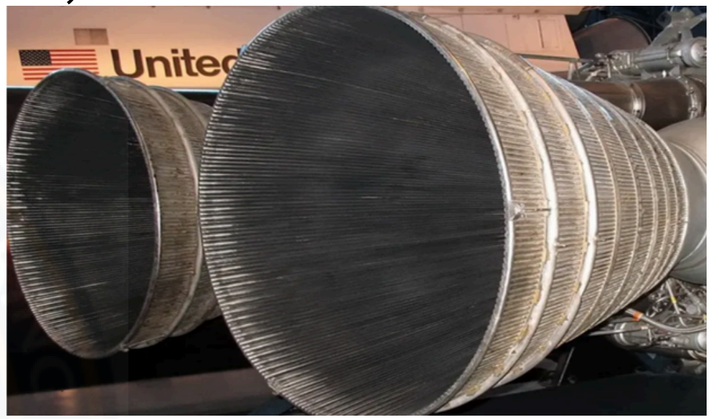
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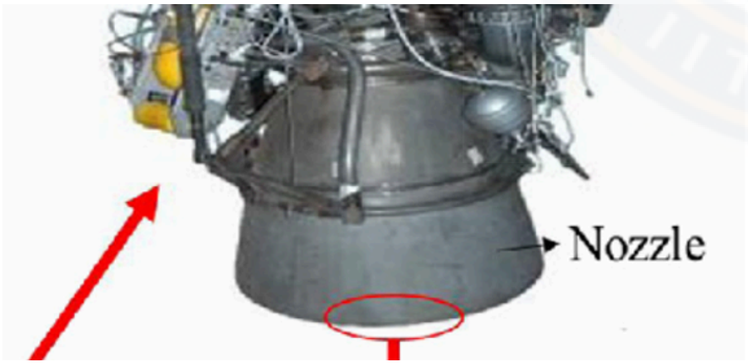
Introduction & Background

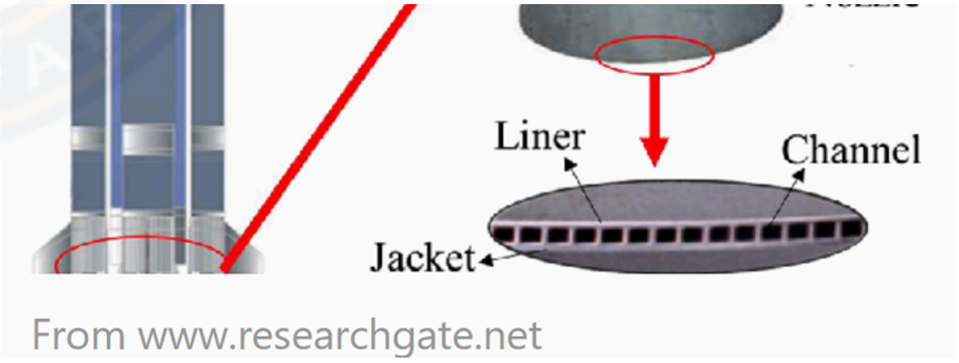
- Modern liquid rocket engines operate under extreme thermal conditions, where combustion gas temperatures exceed 3,000°C.





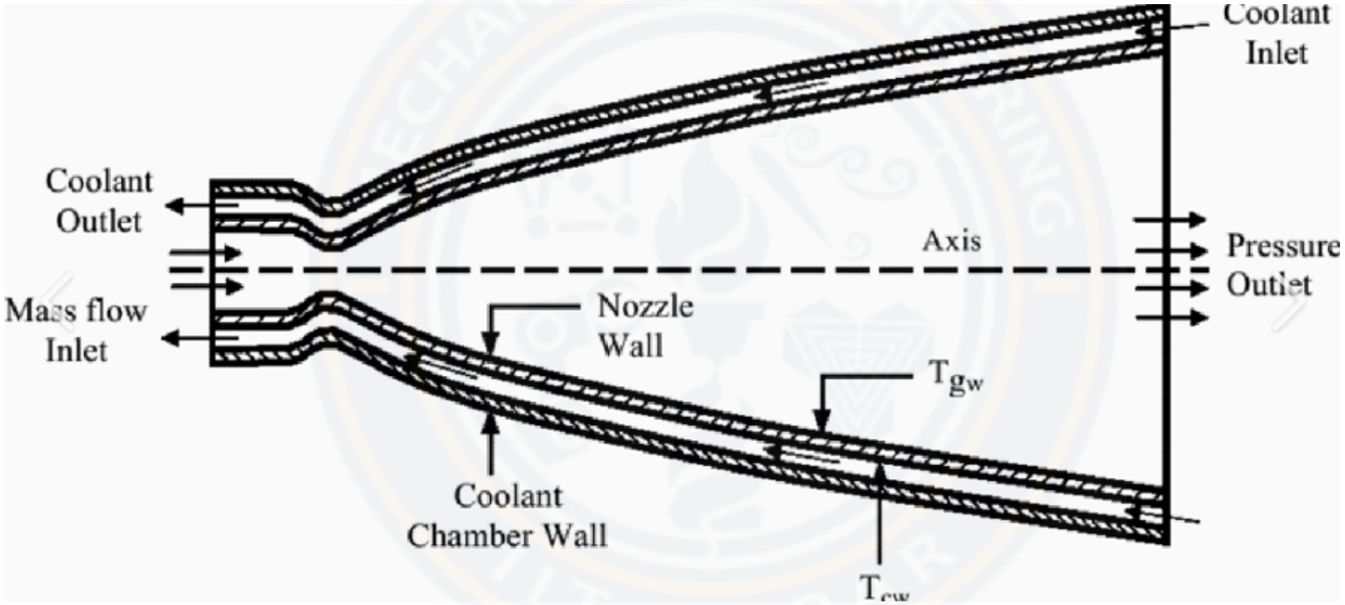
- To prevent the metal chamber walls from melting, engines use regenerative cooling, where cold liquid propellant (fuel or oxidizer) is pumped through internal channels inside the nozzle wall before entering the combustion chamber.





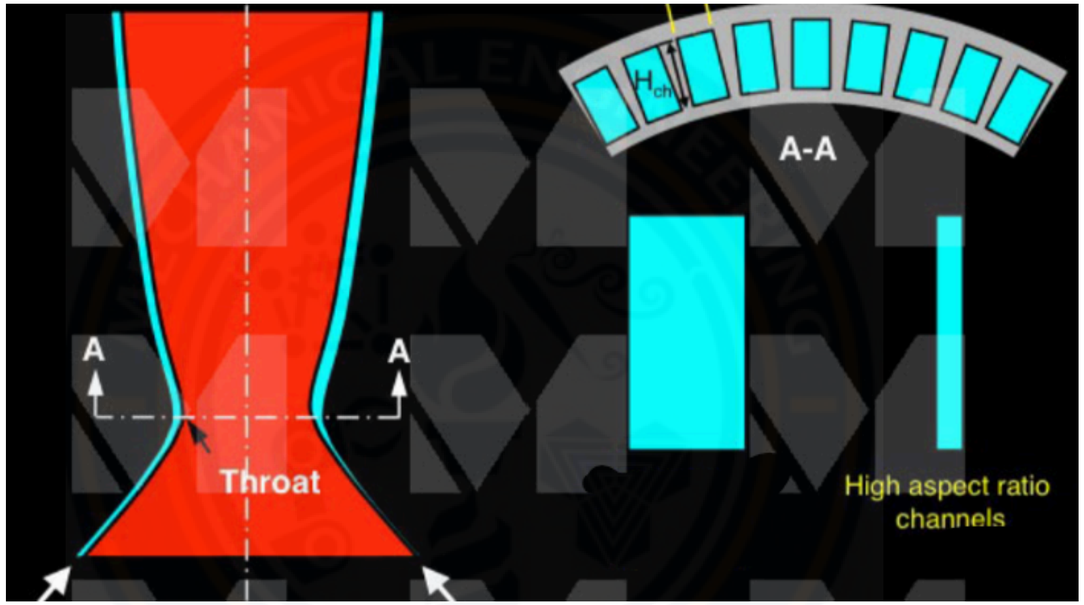
- High thermal conductivity copper alloys (such as CuCrZr) are used for the inner liner, sealed by an electroformed nickel outer jacket.

- Cold propellant enters the outer jacket, flows through narrow cooling passages along the nozzle wall, absorbs extreme heat, and cools the inner surface.



- The heat flux reaches its extreme peak at the nozzle throat—the narrowest section of the engine where gas velocity reaches Mach 1

- To increase cooling performance at the throat, engines use High-Aspect-Ratio Cooling Passages (HARCP), which are deep, narrow rectangular channels designed to boost coolant flow velocity



PAPER REVIEWS

Ref. & Paper Name	Author & Year	Method & Parameters	Key Finding & Limitation
Paper 1: Experimental lifetime study of regeneratively cooled rocket chamber walls	**Felix Hötte et al. (2020)**	**High-heat flux thermomechanical cyclic loading to simulate engine chamber wall environments.**	• **Key Finding:** Confirmed that specimens collapse in the "doghouse failure" mode after several load cycles due to yielding, creep, ratcheting, and thermal ageing. • **Limitation:** Observed and documented the failure experimentally, but did not propose cooling-channel geometry optimization to reduce it.
Paper 2: Integrated Engine-Cycle and Structural Modelling for Reusable Liquid Rocket Engines	**N/A (Integrated Reusable Engine Study)**	**Coupled thermomechanical FEA modeling cyclic plastic deformation under engine operating conditions.**	• **Key Finding**: Identified cyclic plastic deformation at the throat as the primary cause of doghouse distortion, localized bending, and ligament rupture • **Limitation**: Identified doghouse effect as a critical degradation mechanism in reusable engines, but lacked a structural geometric design solution

RESEARCH GAP

What Literature Has Established

- **Experimental Proof: Hötte et al. (2020) experimentally confirmed that conventional flat-walled rectangular channels suffer from thermal plastic ratcheting, leading directly to the Doghouse Effect and low-cycle fatigue failure at the throat.**

- **Paper 2 : Identified cyclic plastic deformation at the throat as the primary cause of doghouse distortion, localized bending, and ligament rupture**

INTELLECTUAL PROPERTY

The Critical Research Gap

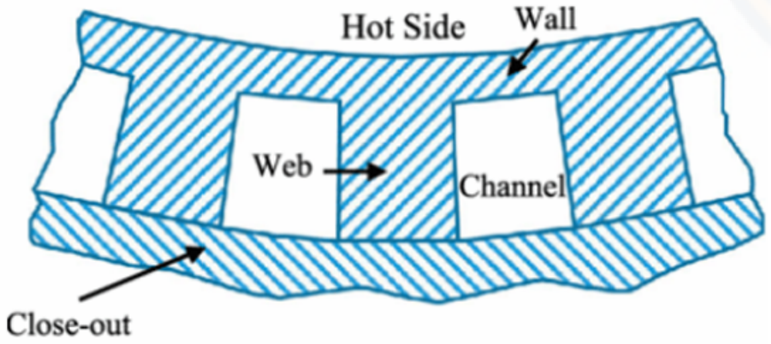
"Existing literature focuses heavily on observing, modeling, and predicting Doghouse failure in standard rectangular channels, but lacks active structural geometry optimization to passively reduce or delay thermal plastic ratcheting at the nozzle throat."

Problem Statement: The Doghouse Effect

- In reusable engines (such as SpaceX Raptor, Blue Origin BE-4, or NASA RS-25), repeated thermal cycling leads to a critical failure mode known as the Doghouse Effect (or thermal plastic ratcheting).

Thermomechanical Failure Sequence:

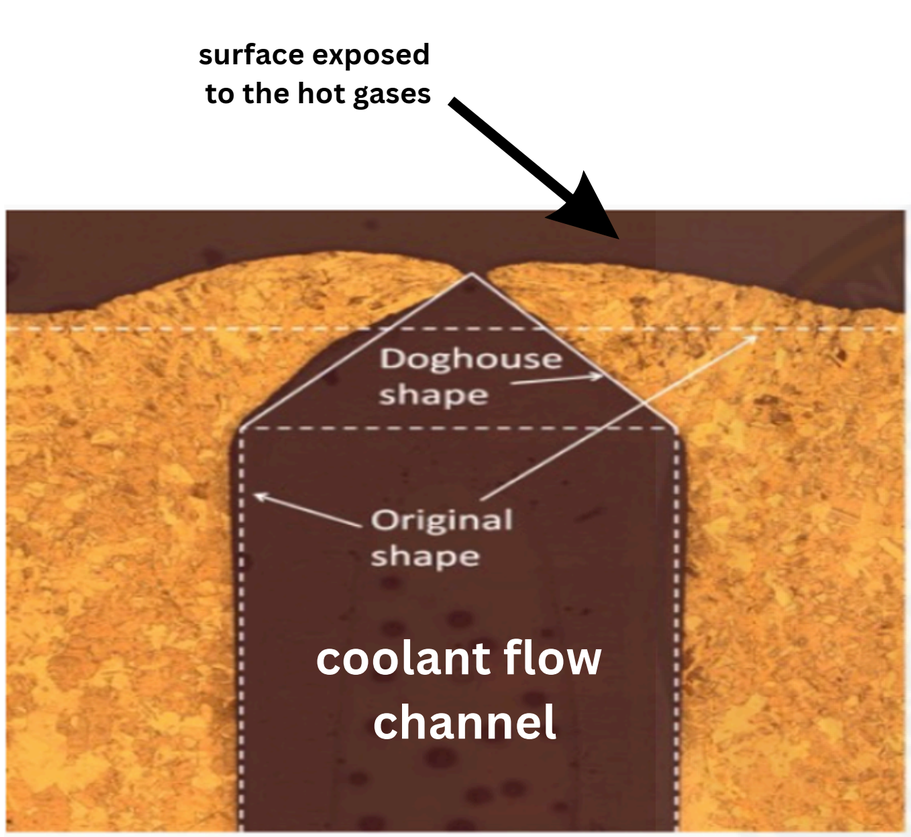
- **Firing Phase :** The hot gas wall experiences intense thermal expansion, but is rigidly constrained by the cooler side ribs and outer jacket. This generates extreme compressive thermal stress exceeding the copper yield limit, causing plastic deformation (thinning/bulging)
- **Shutdown Phase:** Upon cooling, the permanently compressed wall goes into high tensile stress.
- **Cyclic Ratcheting:** Over multiple engine firings, the inner wall progressively bows into the coolant passage, turning the original rectangular profile into a pentagonal "Doghouse" shape.
- **Failure:** Severe wall thinning leads to low-cycle fatigue (LCF) pinhole cracking, coolant leakage, and catastrophic burn-through.

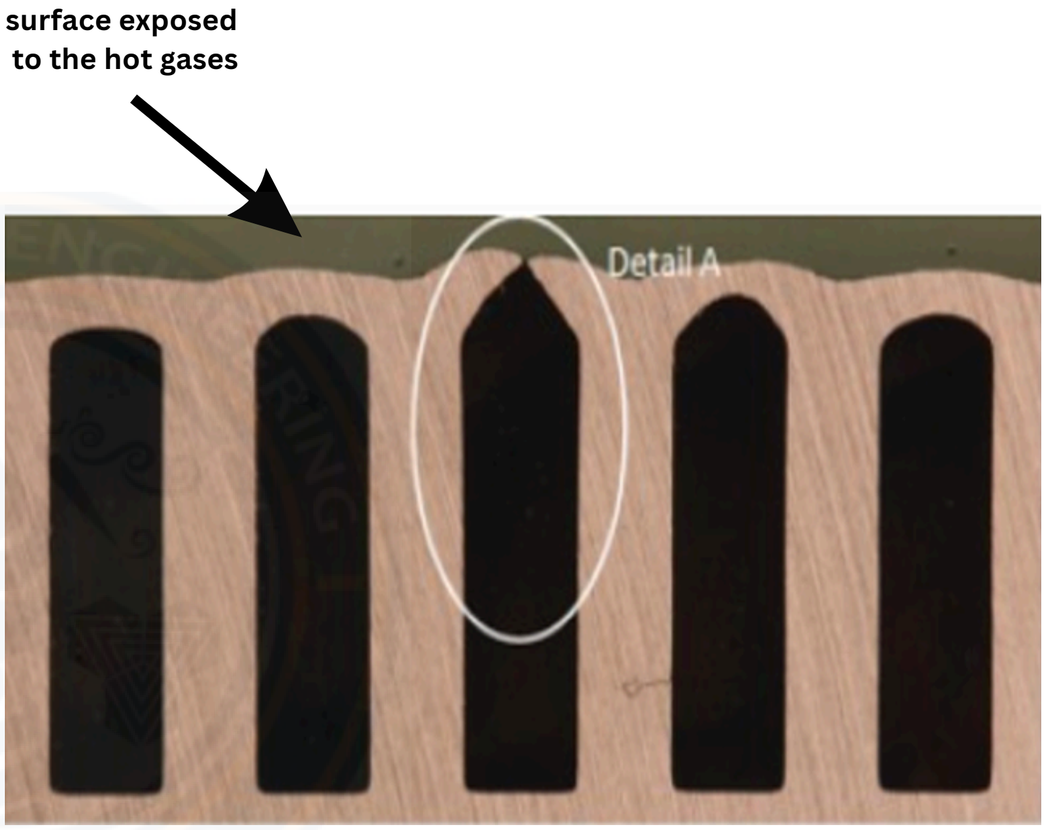




Rupture of Channel Wall

Problem Statement: The Doghouse Effect





THE COOLING CHANNELS BULGES OUTWARD ON THE HOTTER SIDE FROM THE COOLER SIDE

METHODOLOGY

An Innovative Idea to be tested (not yet proved\tested to be effective)

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AIM of the IDEA

This idea aims to increase the life of reusable rocket engines

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Proposed Solution

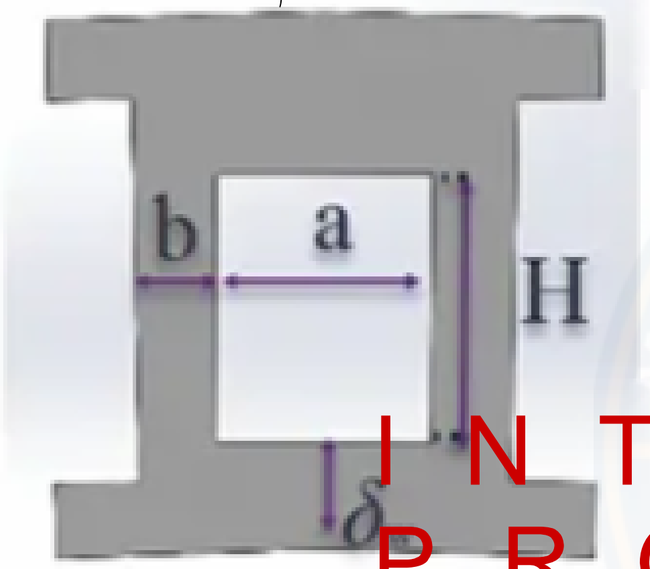
Pre-Domed Curved Geometry of the inner side of the cooling channels

- **To solve this reusability bottleneck, we propose modifying the channel geometry specifically in the nozzle throat region where the Doghouse Effect occurs. Rather than maintaining a flat inner wall with sharp 90° corners, the floor of the channel is manufactured with a smooth, pre-domed (convex) internal contour.**

METHODOLOGY

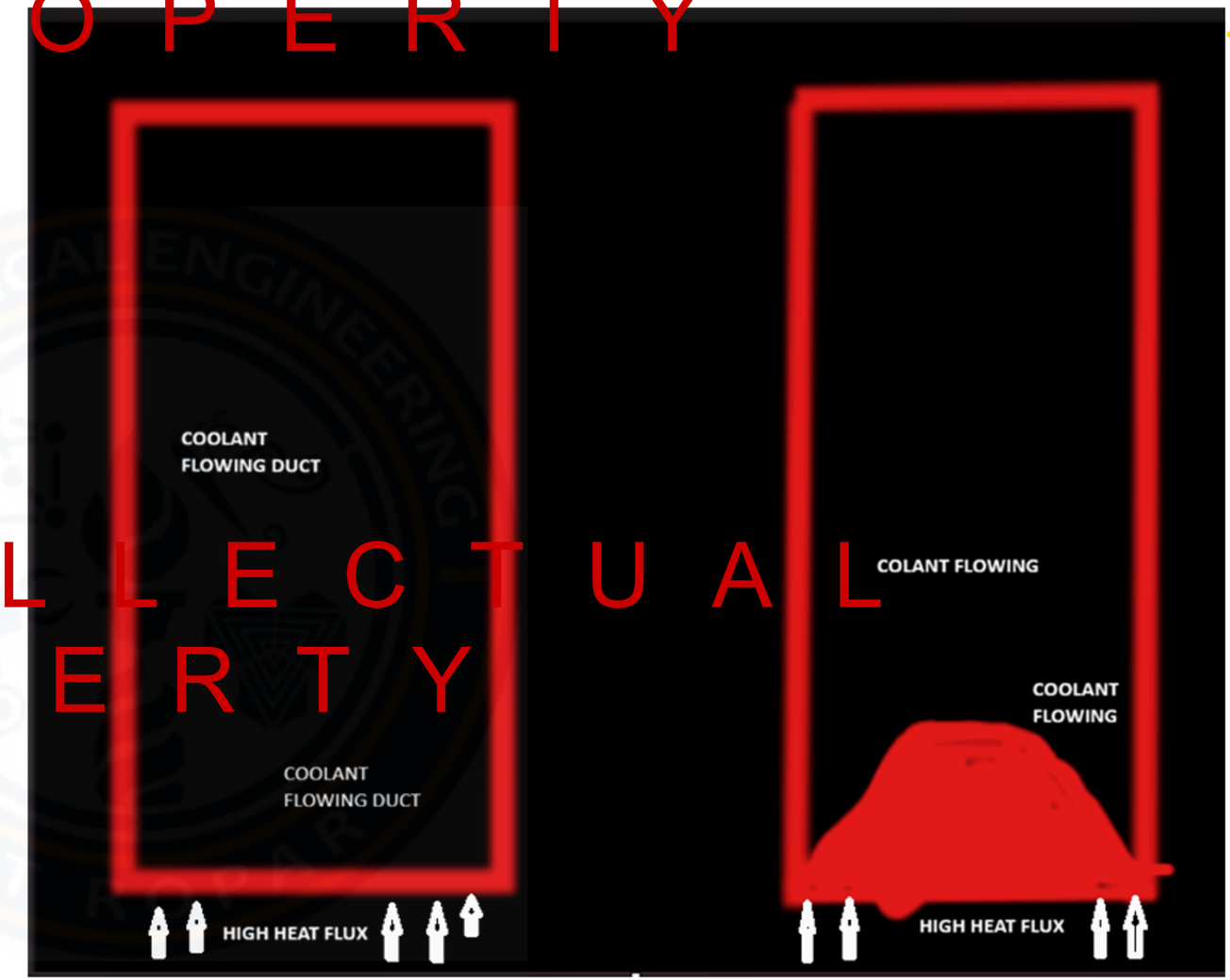
INTELLECTUAL PROPERTY

Outermost side of the nozzle visible to us



The cooling chaannel side exposed to the hot gases

INTELLECTUAL PROPERTY

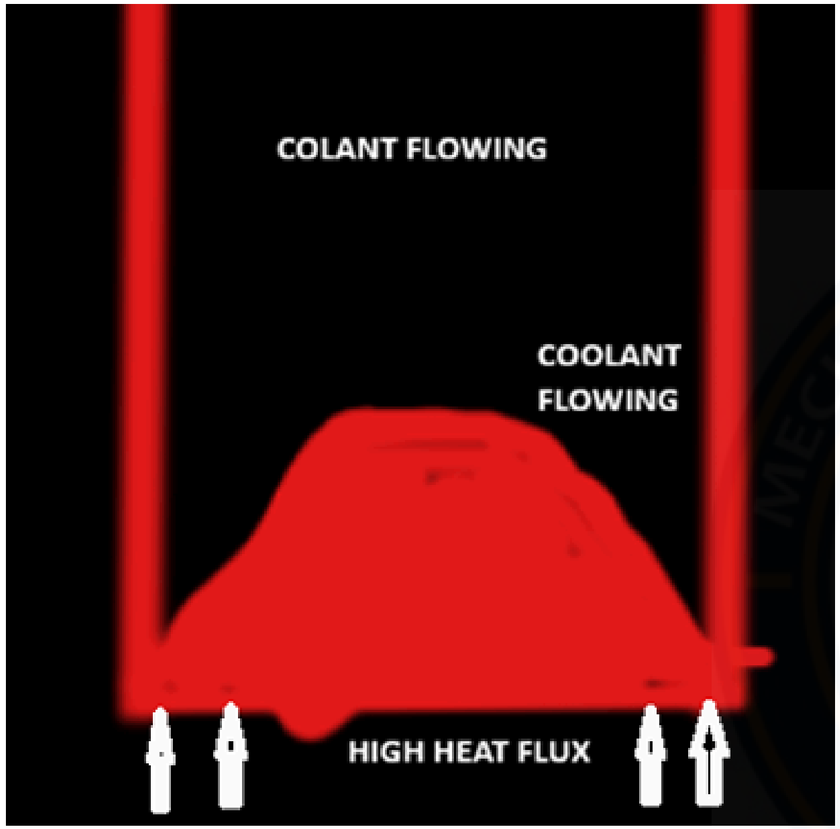


1ST DUCT IS THE CONVENTIONAL ONE
2ND DUCT IS THE PROPOSED ONE

This is the cooling channelduct used in regular regeneratively cooled (reusable also) rocket engine nozzles

METHODOLOGY

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Pre-Domed Curved Geometry of the inner side of the cooling channels

Mechanism of the Proposed Duct

- **Micro-Arch Structural Effect:** The pre-formed parabolic/domed curvature acts as a structural arch. Under intense heating, thermal expansion is converted into uniform hoop compression across the curve, preventing localized plastic buckling into the channel over multiple cycles .
- **Elimination of Corner Stress Risers**: Removing sharp 90° interior corners eliminates high mechanical stress concentration points, spreading fatigue loads smoothly into the side ribs.
- **Enhanced Heat Transfer :** The local constriction directly over the hottest wall region increases local coolant velocity and enhance local heat extraction where heat flux peaks.

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METHODOLOGY

Using thermal and structural simulations, we will try to prove the below :

Parameter	Standard cooling channel geometry	Proposed geometry of the cooling channel
Themal Fatigue resistance	Poor (subject to ratcheting)	Superior (Absorbs thermal strain better)
Structural stress concn	High (sharp 90 deg corner)	Minimal (smooth radius)
Local heat removal	Baseline	Enhanced (more surface for heat transfer)
Pressure drop	Lowest	Moderately inceased
Nozzle Engine Reusability	Usual no.of Cycles	Some Increased no. of cycles

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- **Targeted Application:** Because the rounded channel incurs a slight pumping pressure drop penalty, it is placed exclusively at the high-heat throat region (a length of a few centimeters), leaving the main chamber and divergent nozzle with standard rectangular channels.
- **Manufacturing Feasibility:** Modern 3D metal printing (Laser Powder Bed Fusion - LPBF) with copper alloys easily allows printing complex internal channel geometries without traditional tool access limitations.

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INTELLECTUAL PROPERTY

Conclusion

The proposed pre-domed channel geometry is expected to effectively mitigates plastic ratcheting and doghouse failure at the throat, significantly enhancing structural integrity and engine reusability for modern space launch systems.


THANK YOU